INFRAFLOW

Discovery Brief

Product: Infrastructure Migration Command Center

Document version: 1.0

Project type: Independent product case study

Status: Concept and discovery hypothesis

Prepared by: Product / Project Management

1. Product Overview

InfraFlow is a proposed centralized workspace for planning, monitoring, and governing IT infrastructure migrations.

It aims to help project managers, PMO analysts, business analysts, and technical leads coordinate application dependencies, migration milestones, project risks, and change approvals through a consistent source of project information.

The product is intended to improve operational visibility and support better-informed project decisions.

2. Problem Statement

Infrastructure migration programs can involve interconnected applications, multiple technical teams, competing timelines, operational risks, and approval dependencies.

When migration information is maintained across disconnected spreadsheets, status reports, and communication channels, stakeholders may face challenges in understanding dependencies, identifying unresolved risks, tracking delivery progress, and maintaining consistent reporting.

How might we help IT migration teams coordinate dependencies, monitor delivery risks, and maintain governance visibility through a centralized migration management workspace?

This is a problem hypothesis to be investigated, not a validated finding.

3. Target Users

Primary users

- **Project Manager:** Coordinates delivery plans, milestones, task owners, blockers, and escalations.
- **PMO Analyst:** Maintains governance records, monitors project health, and prepares status reports.
- **Technical Lead:** Identifies application dependencies, assesses technical constraints, and coordinates migration readiness.

Secondary users

- **Business Analyst:** Documents requirements, maps business and technical dependencies, and maintains traceability.
- **Executive Sponsor:** Reviews project health, major risks, schedule performance, and decisions requiring escalation.

4. Stakeholder Needs

Stakeholder	Primary need
Project Manager	One view of progress, milestones, blockers, and risks
PMO Analyst	Consistent reporting and traceable governance decisions
Technical Lead	Visibility into application relationships and migration sequencing
Business Analyst	Clear requirements, ownership, and dependency records
Executive Sponsor	Concise status reporting and visibility into critical issues

5. Discovery Assumptions

The following assumptions guide the initial product concept:

1. Migration teams need a consistent view of application dependencies and migration status.
2. Project stakeholders benefit from a centralized risk and issue register.
3. Migration milestones require clear ownership, planned dates, and status tracking.
4. Changes to migration plans should follow a documented review and approval process.
5. Different stakeholder groups require different levels of project detail.

6. A focused minimum viable product can demonstrate the value of connecting these workflows.

These assumptions must be validated through stakeholder interviews, workflow walkthroughs, and feedback on an initial prototype.

6. Proposed Discovery Questions

The following questions would be used in future stakeholder interviews:

- How is the current migration plan maintained and updated?
- Where are application dependencies documented?
- How do teams identify and escalate migration risks?
- What information is required for weekly project status reviews?
- Who can approve changes to migration dates or scope?
- Which reporting tasks require the most manual effort?
- What information is missing when stakeholders make migration decisions?

7. Product Objectives

InfraFlow aims to:

1. Centralize migration planning information.
2. Improve visibility into application dependencies.
3. Standardize risk ownership and mitigation tracking.
4. Make milestone progress and overdue activities visible.
5. Establish a traceable change-request workflow.
6. Support consistent project reporting and escalation.

8. MVP Scope

In scope

- Application inventory and migration status.
- Application dependency mapping.
- Migration milestones and timeline.
- Risk and issue register.
- Executive project dashboard.
- Change requests and approval tracking.
- Basic reporting and record history.

Out of scope

- Direct cloud-provider integrations.

- Automated infrastructure discovery.
- AI-driven risk prediction.
- Actual migration execution or deployment.
- Enterprise single sign-on and advanced permissions.
- Financial forecasting and automated cost optimization.

9. Proposed Success Measures

Success will be evaluated against defined criteria once a representative prototype or MVP exists.

Metric	Measurement approach
Dependency coverage	Percentage of in-scope applications with documented dependencies
Ownership coverage	Percentage of active risks with assigned owners
Milestone visibility	Percentage of milestones with owners, dates, and status
Change traceability	Percentage of change requests with documented decisions
Reporting effort	Time required to prepare a standardized status report

Baselines and targets have not yet been established. No business impact or improvement is claimed at this stage.

10. Discovery Outcome

The discovery phase establishes a working problem hypothesis, identifies the main stakeholder groups, defines assumptions requiring validation, and sets boundaries for the initial product scope.

The next phase translates these findings into prioritized requirements, user stories, and acceptance criteria.

Product Requirements Document (PRD)

1. Purpose

This document defines the initial product requirements for InfraFlow, a proposed platform for coordinating IT infrastructure migration activities, monitoring project health, and maintaining governance records.

It establishes the functional scope, user needs, acceptance criteria, and non-functional considerations for the initial MVP.

2. Product Goals

The MVP should enable users to:

- Maintain a centralized application inventory.
- Document relationships between applications.
- Plan and monitor migration milestones.
- Record, prioritize, and track risks.
- Submit and review change requests.
- Monitor project health through a consolidated dashboard.

3. User Roles

Role	Core responsibility
Project Manager	Manage milestones, risks, and delivery status
PMO Analyst	Monitor governance and prepare reports
Technical Lead	Maintain technical dependencies and readiness information
Business Analyst	Support requirements and dependency traceability
Executive Sponsor	Review project health and escalations

These are conceptual roles for the MVP. A production system would require a more detailed permissions model.

4. Functional Requirements

Priorities use MoSCoW: Must have, Should have, Could have, and Won't have in the current release.

ID	Requirement	Priority
FR-01	Users can create and update application records.	Must
FR-02	Each application can have an owner, criticality, migration status, and migration wave.	Must
FR-03	Users can document upstream and downstream application dependencies.	Must
FR-04	Users can view application relationships in a dependency map.	Must
FR-05	Users can create and update migration milestones with owners, dates, and statuses.	Must
FR-06	Users can record risks with probability, impact, owner, mitigation, due date, and status.	Must
FR-07	The dashboard displays migration progress, upcoming milestones, overdue milestones, open risks, and pending approvals.	Must
FR-08	Users can submit change requests with a reason, affected records, impact, and proposed change.	Must
FR-09	Authorized reviewers can approve or reject change requests with a recorded decision.	Must
FR-10	Users can generate a standardized project status summary.	Should
FR-11	The system maintains a history of important changes and approval decisions.	Should
FR-12	Users can filter records by status, owner, migration wave, and severity where relevant.	Should

5. User Stories and Acceptance Criteria

US-01: Application Inventory

As a technical lead,

I want to maintain application records and migration status,

so that the project team has a consistent view of the migration scope.

Acceptance criteria:

- A user can create an application with a unique identifier and name.

- Each application can include an owner, criticality, migration status, and migration wave.
- Required fields are validated before saving.
- The application can be located through search or filtering.

US-02: Dependency Mapping

As a technical lead,

I want to view upstream and downstream application dependencies,

so that I can identify sequencing constraints before a migration wave begins.

Acceptance criteria:

- Users can link two existing application records.
- Each dependency identifies the related applications and relationship.
- The map visually distinguishes applications and their connections.
- Selecting an application reveals its documented dependencies.
- Invalid self-referential dependencies are rejected.

US-03: Risk Management

As a project manager,

I want to assign owners and mitigation actions to migration risks,

so that unresolved risks can be monitored and escalated.

Acceptance criteria:

- A risk record includes a description, probability, impact, owner, mitigation, due date, and status.
- Risks can be filtered by severity, owner, and status.
- High-severity risks without an owner are visibly flagged.
- Closed risks remain available for historical reference.

US-04: Milestone Tracking

As a project manager,

I want to monitor milestone dates, ownership, and status,

so that I can identify delivery slippage.

Acceptance criteria:

- Each milestone includes a name, owner, planned date, and status.
- Incomplete milestones past their planned dates are flagged as overdue.
- Users can view upcoming and overdue milestones.
- The dashboard reflects the latest saved milestone status.

US-05: Executive Dashboard

As an executive sponsor,

I want to review project health in a consolidated dashboard,

so that I can identify critical risks and decisions requiring attention.

Acceptance criteria:

- The dashboard displays migration progress and milestone status.
- It highlights overdue milestones, high-severity risks, and pending approvals.
- Each KPI has a clear definition and consistent calculation.
- Selecting a summary metric leads to the relevant underlying records, where supported.

US-06: Change Approval

As a project manager,

I want to submit proposed changes for review,

so that changes to migration plans are assessed and documented before implementation.

Acceptance criteria:

- A change request records the reason, proposed change, affected items, submitter, and submission date.
- Each request has a status: Draft, Submitted, Approved, Rejected, or Withdrawn.
- Only an authorized reviewer can approve or reject a submitted request.
- Approval or rejection requires a recorded decision and timestamp.
- Rejected requests retain their decision history.
- Approval of a request does not automatically mean the underlying change has been implemented.

6. Non-Functional Requirements

These are proposed design requirements for the MVP, not verified system capabilities.

Usability

- Primary workflows should be understandable without extensive training.
- Status labels and risk severity indicators should be consistent across screens.
- Important actions should provide clear confirmation or validation feedback.

Data integrity

- Each application, risk, milestone, and change request should have a unique identifier.
- Required fields and permitted status transitions should be validated.
- Dashboard metrics should reconcile with the underlying records.

Security and access

- A future implementation should enforce role-based access to sensitive records and approval actions.
- Approval decisions and important changes should be attributable to a user.
- Production deployment would require appropriate authentication, authorization, and data-protection controls.

Performance

- Proposed MVP target: primary dashboard and record views should load within three seconds under a defined representative test dataset and environment.
- This target must be validated through performance testing.

7. Business Rules

1. Every application must have a unique identifier.
2. A dependency can only reference existing application records.
3. Every active risk must have a status; high-severity risks without owners must be flagged.
4. An incomplete milestone past its planned date is considered overdue.
5. A change request must be submitted before it can be approved or rejected.
6. Only authorized reviewers may record approval decisions.
7. Approved changes must be tracked separately from the execution of those changes.
8. Dashboard KPIs must use documented definitions and consistent source data.

8. MVP Prioritization

Must have: Application inventory, dependency mapping, milestones, risk register, executive dashboard, change approval.

Should have: Standardized reporting, change history, advanced filters.

Could have in a future release: Notifications, collaboration features, additional analytics, external integrations.

Won't have in the MVP: Automated cloud discovery, AI risk prediction, infrastructure deployment, and advanced enterprise integrations.

9. Dependencies and Constraints

- Representative application and migration data must be available for testing.
- Stakeholders must agree on status definitions and risk severity rules.
- Dependency relationships must be documented accurately by the relevant technical owners.
- Approval responsibilities must be defined before implementing change control.
- The initial case study assumes a limited, single-program MVP rather than a full enterprise deployment.

10. Acceptance and Evaluation Plan

The proposed MVP will be evaluated through test scenarios covering:

- Creating and updating application records.
- Visualizing valid application dependencies.
- Identifying overdue milestones.
- Flagging high-severity risks without owners.
- Reconciling dashboard KPIs against underlying records.
- Submitting, approving, and rejecting change requests.
- Confirming that approval decisions retain a traceable history.

A requirement will be considered satisfied only when its acceptance criteria have been demonstrated and documented.

11. Open Questions

The following questions require further discovery:

- Which existing tools currently hold migration data?
- What is the expected scale of applications and migration waves?
- How are risk severity and project health calculated today?
- Which changes require formal approval?
- What access restrictions apply to technical and executive stakeholders?
- What reporting formats are required by the PMO?

12. Definition of Done

The requirements phase is complete when:

- The MVP scope and exclusions are documented.

- Functional and non-functional requirements are defined.
- User stories have testable acceptance criteria.
- Priorities and dependencies are recorded.
- Open assumptions and discovery questions are visible.
- Requirements are ready for review, prototyping, and implementation planning.
